

The Positive Defect Problem: Target and Admissibility Criteria for a Programmatic Search for Unforced Navier-Stokes Blowup

J. Petrillo
jarret@homology.io

J. Glimm
Stony Brook University, Stony Brook, NY 11794, and GlimmAnalytics LLC, USA
glimm@ams.sunysb.edu

September 20, 2026

Abstract

On 7–8 September 2026 programmatic search produced singularities: a forced Navier-Stokes singularity at every fixed viscosity — the breakdown statements (C) and (D) of the Clay problem — and two Euler singularities [27, 28, 1]. The unforced problem, statements (A) and (B), stands open, and a search for it needs a target. This paper fixes one: the positive defect problem, that a Leray-Hopf solution from smooth data on the periodic cube loses energy on a finite window, at fixed viscosity, beyond what viscosity removes. A positive defect implies blowup and so a negative answer to statement (B); the converse is not known, and the target is at least as strong as the problem it addresses. The paper proves the target equivalent to a floor on the energy flux through the Littlewood-Paley shells — the Fourier-side form of the coarse-grained flux Π_ℓ of the Onsager theory of turbulence — averaged over the window; states necessary conditions on a candidate — a singular time of Type II in velocity, energy concentrating on a set of zero length, a pressure outside L^2 , a velocity outside the Onsager-critical class $L_t^3 B_{3,c_0}^{1/3}$, an obstruction to collapse onto a fixed steady Euler profile; and states what cannot certify one — no finite computation witnesses a Galerkin-uniform ceiling, and selection and forcing return the question to a positive defect. The authors’ pseudo-spectral search at 128^3 and 256^3 measures which of the two conditions of the reduction is the binding one: the fine-shell flux floor holds to within 1–6% of the ceiling and persists for a third of a turnover time, and fails in scale at the Kolmogorov wavenumber, so a candidate must differ from generic turbulence in the depth of its cascade and not in its timing. Every implication not marked otherwise is a theorem in Lean 4 over Mathlib; the library contains no Navier-Stokes object, and the equation enters only through hypotheses.

1 Introduction

On 8 September 2026 OpenAI announced a finite time singularity for the forced three-dimensional Navier-Stokes equations at every fixed viscosity, statements (C) and (D) of the Clay formulation [16], and an unforced finite time singularity for the three-dimensional Euler equations [27, 28]; a preprint of

Alpöge and Buckmaster dated 7 September 2026 constructs an Euler singularity under a smooth force [1]. The announcement states that the constructions and their Lean formalizations were produced by a system of coordinating agents, statements (A)–(D) assigned to separate groups [27]; the papers describe no method.

When solutions are searched for programmatically at this scale, the value of a result shifts to the specification of the target and of the conditions a candidate must satisfy. This paper supplies that specification for one route to the question that remains open, the unforced problem, statements (A) and (B): the positive defect problem of Sec. 2, a sufficient condition for a negative answer to statement (B). The target is one statement: a positive defect, which is what the floor hypothesis of the floor-ceiling argument for finite time loss of regularity (Sec. 2) amounts to once integrated over the window. A machine-checked implication (Theorem 3.2) moves it to the dyadic shells, where it becomes a condition on a quantity a flow exhibits: a positive defect holds exactly when the energy transfer out of every sufficiently fine shell stays above a fixed level on average over the window, and the stronger pointwise floor — the same bound at each instant — implies it. A candidate must satisfy the necessary conditions of Sec. 4, against which the announced constructions are placed; a Galerkin computation, a selection principle and a force cannot certify one (Sec. 5); the same section compares the authors’ own search, twelve pseudo-spectral runs at 128^3 and 256^3 , with the two conditions of the reduction, and finds that a candidate must differ from generic turbulence in the depth of its cascade, not in its timing. Since Tao’s averaged equation [35] blows up with no defect, the floor-ceiling argument is conditional on a statement at least as strong as the regularity problem. Only fluids on $\mathbb{T}^3$ with no external force and fixed viscosity $\nu > 0$ are considered, the setting of statement (B).

Every implication is a theorem in Lean 4 over Mathlib, compiling with no unproven step and no axiom beyond propositional extensionality, choice and quotient soundness. The library contains no Navier-Stokes object: its theorems relate real sequences and real functions of time, and the equation enters only through their hypotheses, classical facts accepted from the literature and a positive defect. Each verified statement carries its declaration name inline, as [L: name], resolving to `EnergyDefect.name` in the library distributed with the paper at https://github.com/researchathomology/NS_ENERGY_DEFECT_REDUCTION (Lean v4.33.0, Mathlib v4.33.0).

Connecting it to the formalized Leray-Hopf theory of [37], so that Sec. 3 is proven for Navier-Stokes objects rather than assumed for real sequences, is not done here.

2 Setting and the Positive Defect Problem

Spaces $\mathcal{H}$ is the space of L^2 divergence free mean zero vector fields on $\mathbb{T}^3$, with inner product $\langle \cdot, \cdot \rangle$; $\mathcal{V} = \mathcal{H} \cap H^1$; $E(t) = \frac{1}{2} \langle u, u \rangle$. A Leray-Hopf (LH) solution satisfies $u \in L^\infty(0, T; \mathcal{H}) \cap L^2(0, T; \mathcal{V})$, is weakly continuous into $\mathcal{H}$, has $\partial_t u \in L^{4/3}(0, T; \mathcal{V}')$, and satisfies the energy inequality $E(t) + \nu \int_s^t \|\nabla u\|^2 \leq E(s)$, $t \geq s$, from almost every initial time s [30]. An initial time s from which that inequality holds is called *good*; almost every time is good, and $s = 0$ is good. The solution is *strong* on an interval if in addition $u \in L^\infty(\mathcal{V}) \cap L^2(H^2)$ there; a strong solution is smooth in the interior of that interval, is the only LH solution from its data on that interval [29, 32], and satisfies the energy identity with equality rather than inequality. The epoch of regularity is the maximal interval $[0, e)$ on which the solution is strong; blowup is $e < \infty$. A time is singular if the solution is not strong on any neighborhood of it, and regular otherwise.

The Energy Defect For $0 \leq w_0 < w$ set

$$\mathcal{A}(w_0, w) = E(w_0) - E(w) - \nu \int_{w_0}^w \|\nabla u\|^2 dt. \quad (1)$$

This is the energy lost on the window beyond what viscosity removes: zero for a strong solution, nonnegative for an LH solution when w_0 is a good time in the sense above. It is the total mass on $\mathbb{T}^3 \times (w_0, w]$ of the inertial dissipation distribution $D(u)$ of Duchon and Robert [14] at fixed ν , the object through which the mathematical literature relates the loss of energy of a weak solution to its regularity, measured in Besov spaces: $D(u)$ vanishes, for Euler and for Navier-Stokes alike, when $u \in L_t^3 B_{3, c_0}^{1/3}$ [14, 9, 10], sharpening the Hölder threshold $1/3$ of Onsager [26, 15, 12], below which dissipative Euler solutions exist [21, 4].

Sec. 3 states the floors of this paper in that language. The defect is not the zeroth law of turbulence, which concerns dissipation surviving as $\nu \rightarrow 0$ along a family of solutions.

The turbulent fluctuation ν_t is the rate of this loss, the inertial dissipation rate: $\int_{w_0}^w \nu_t dt = \mathcal{A}(w_0, w)$ on every window, and $\nu_t = 0$ for a strong solution. A floor is a lower bound and a ceiling an upper bound on a rate; neither word denotes a limit.

The Floor-Ceiling Argument The argument is arithmetic, and it is a statement about a rate rather than about a solution [L: epoch_le_of_decayBounds, le_crossingTime, epoch_le_of_shifted_floor_polynomial_ceiling]. Let $r \geq 0$ be a rate defined on the epoch of regularity $[0, e)$ of an LH solution. If $A, B > 0$ and $b < a$ satisfy the floor $Be^{-bt} \leq r$ and the ceiling $r \leq Ae^{-at}$ on $[0, e)$, then $e \leq \max(0, \log(A/B)/(a-b))$; if $B, q > 0$ and $w_0 \geq 0$ satisfy the constant floor $B \leq r$ and the polynomial ceiling $r \leq Ct^{-q}$ on $[w_0, e)$, then $e \leq \max(w_0, (C/B)^{1/q})$. A short epoch follows from the two bounds alone. What the argument does not supply is a rate that carries them.

The choice of r decides what the floor asserts. Taking $r = \nu_t$ makes it a positive defect outright: $B \leq \nu_t$ on a window integrates to $\mathcal{A} \geq B\ell > 0$ for ℓ the length of the window, so the hypothesis already states the loss of regularity that the conclusion reports; and since ν_t vanishes wherever the solution is strong, that floor cannot hold on $[w_0, e)$ for any $e > w_0$. The floor must therefore be carried by a rate that a strong solution can exhibit. The shell flux of Sec. 3 is such a rate, and is the one the validated computation of Sec. 5 bounds. Neither bound is derived here; whether some LH solution from turbulent C^∞ data admits such a pair is not addressed.

The positive defect problem. There exist C^∞ divergence free data u^0 on $\mathbb{T}^3$ and times $0 \leq w_0 < w$ such that a Leray-Hopf solution of the Navier-Stokes equation from u^0 satisfies

$$\mathcal{A}(w_0, w) = E(w_0) - E(w) - \nu \int_{w_0}^w \|\nabla u\|^2 dt > 0.$$

Status: open; the forced singularity of [27] lies outside

the unforced class (Sec. 4). This is question (a) of Duchon and Robert [14] — does a weak solution in $L_t^2 H_x^1 \cap L_t^\infty L_x^2$ with nonzero inertial dissipation exist — restricted to smooth data; it is open for general L^2 data as well. The phenomenology of turbulence gives no reason to expect a positive answer. A positive defect requires energy to be transferred through the cascade to arbitrarily fine scales and lost there; but at a fixed viscosity the cascade stops at the Kolmogorov scale, and below that scale viscosity dissipates the whole of the incoming flux, so nothing is left to pass to the limit. That is evidence against a positive defect at fixed ν , not a proof that none occurs, and the measurements of Sec. 5 are a quantitative version of the same statement.

3 The Reduction

This section restates a positive defect as a condition on the energy transfer between scales. It sets up the dyadic shells and the flux through them, records the energy balance of a truncation and its limit (Lemma 3.1), states two shell floors — one pointwise in time at every fine scale, one averaged over the window — and proves that the averaged floor is exactly a positive defect, that the pointwise floor implies it, and that either one forces blowup (Theorem 3.2). The averaged shell floor is the target of the paper. The whole content is the energy balance of a Fourier truncation and its limit.

Let $\mathbb{P}$ denote the Leray projection onto divergence free fields and $b(u, v, w) = \langle u \cdot \nabla v, w \rangle$ the trilinear form of the nonlinearity. For $j \geq 0$ the shell S_j is the set of Fourier modes with $2^j \leq |k| < 2^{j+1}$, $\Delta_j u$ the part of u on S_j , $\mathbb{P}_{\leq j} = \sum_{i \leq j} \Delta_i$ the projection onto the modes up to shell j , and $E_{\leq j} = \frac{1}{2} \|\mathbb{P}_{\leq j} u\|^2$ the energy they hold. The flux across the boundary of shell j is

$$\Pi_j = \langle \mathbb{P}_{\leq j} u, \mathbb{P}(u \cdot \nabla u) \rangle = b(u, u, \mathbb{P}_{\leq j} u), \quad (2)$$

so that $\frac{d}{dt} E_{\leq j} = -\Pi_j - \nu \|\nabla \mathbb{P}_{\leq j} u\|^2$; the sign is fixed by this balance.¹

¹This is the convention of [34]; [9] take the opposite sign.

The Same Objects in Standard Language The shells are the dyadic blocks of Littlewood-Paley theory, with a sharp Fourier cutoff in place of the usual smooth one, and Δ_j is the corresponding block. With the smooth blocks Δ_j^{sm} the Besov norm $\|u\|_{B_{p,q}^s}$ is the ℓ^q norm of the sequence $2^{js}\|\Delta_j^{\text{sm}}u\|_{L^p}$, and B_{p,c_0}^s is the subspace on which that sequence tends to zero; the sharp blocks give the same norm for $p = 2$ and not for $p \neq 2$, where the sharp projector is unbounded on L^p . The sharp cutoff costs nothing here, since only L^2 quantities enter Lemma 3.1 and Theorem 3.2, and it is what makes $\mathbb{P}_{\leq j}u$ a finite-mode test field in the proof of the lemma; the Besov classes named in this paper are the standard ones, defined by the smooth blocks. The flux Π_j is the Fourier-side form of the coarse-grained flux of the Onsager theory of turbulence [15, 12, 14]. There one averages over the length scales ℓ and finer by mollifying, $u_\ell = \varphi_\ell * u$, and the balance for the resolved energy $\frac{1}{2}|u_\ell|^2$ carries the flux through scale ℓ , $\Pi_\ell = -\nabla u_\ell : \tau_\ell$, with $\tau_\ell = (u \otimes u)_\ell - u_\ell \otimes u_\ell$ the subscale stress; integrated over $\mathbb{T}^3$ it is $\langle \varphi_\ell * u_\ell, u \cdot \nabla u \rangle$. Replacing the mollifier by the projector $\mathbb{P}_{\leq j}$, for which $\mathbb{P}_{\leq j}^2 = \mathbb{P}_{\leq j}$, at $\ell \sim 2^{-j}$ turns this into (2) exactly. Lemma 3.1 below is the sharp-cutoff form of the Duchon-Robert balance integrated over space, and its limit $j \rightarrow \infty$ is the statement that $D(u)$ has total mass $\mathcal{A}(w_0, w)$ on the window, Proposition 1 of [14] without its localization in space — the filter differs, the limit does not; the localized form is used in Sec. 4. The floors stated after the lemma are, in this language, lower bounds on Π_ℓ uniform in ℓ below some ℓ_0 , pointwise in time or averaged over the window.

Lemma 3.1 (Shell balance) [L: shell_partial_sums_tendsto, shell_dissipation_tendsto, flux_tendsto_defect] For an LH solution and $0 \leq w_0 < w$,

$$F_j := \int_{w_0}^w \Pi_j dt = E_{\leq j}(w_0) - E_{\leq j}(w) - \nu \int_{w_0}^w \|\nabla \mathbb{P}_{\leq j}u\|^2 dt, \quad (3)$$

and $F_j \rightarrow \mathcal{A}(w_0, w)$ as $j \rightarrow \infty$.

Proof. $\mathbb{P}_{\leq j}u$ has finitely many Fourier modes, so it is a legitimate test field: $\langle \partial_t u, \mathbb{P}_{\leq j}u \rangle$ is an $L^{4/3}$ -

L^∞ pairing and $t \mapsto \|\mathbb{P}_{\leq j}u\|^2$ is absolutely continuous with endpoint values given by weak continuity; integrating gives (3). As $j \rightarrow \infty$, $E_{\leq j}(t) \uparrow E(t)$ by Parseval and the coarse dissipation increases to $\nu \int \|\nabla u\|^2 < \infty$ by monotone convergence, using $u \in L^2(0, T; \mathcal{V})$. $\square$

Let $\ell = w - w_0$. By Sec. 2 the floor of the argument has to be carried by a rate a strong solution can exhibit; the shell flux is such a rate, and the reduction uses two floors on it. The *pointwise floor* is: there are j_0 and $B > 0$ with $\Pi_j(t) \geq B$ for all $j \geq j_0$ and all $t \in [w_0, w]$. The *averaged floor* is: there are j_0 and $B > 0$ with $F_j \geq B\ell$ for all $j \geq j_0$. Both are uniform in the scale, and that is the essential feature of each: a floor on Π_j at one fixed j is compatible with regularity, whereas the same bound holding at every $j \geq j_0$ with one constant B is not. They differ in the other variable. The pointwise floor asks for the bound at each instant of the window; the averaged floor asks only that the window integral F_j stay above $B\ell$, and is the weaker of the two.

The two floors and the defect are ordered:

$$\begin{aligned} \mathcal{A}(w_0, w) > 0 &\iff \text{averaged floor} , \\ \text{pointwise floor} &\implies \text{averaged floor} . \end{aligned}$$

The converse of the second is not claimed. Both are Theorem 3.2. The averaged floor is equivalent to the defect and weaker than the pointwise floor, and it is the target of this paper: it is the least that has to be proved, and it is stated in terms of the transfer between scales, which a candidate flow can be asked to exhibit.

Theorem 3.2 (The averaged floor is a positive defect) [L: averaged_tail_floor_le_defect, defect_gives_averaged_tail_floor, energy_equality_no_averaged_tail_floor, pointwise_tail_floor_le_defect] For an LH solution and a window $[w_0, w]$: (a) the averaged floor holds iff $\mathcal{A}(w_0, w) > 0$; (b) the pointwise floor implies the averaged floor, hence $\mathcal{A}(w_0, w) \geq B\ell > 0$; (c) in either case the solution is not strong on $[w_0, w]$, and by weak-strong uniqueness [29, 32] the strong solution from the same data has blown up by time w .

Proof. (a) If $F_j \geq B\ell$ for all $j \geq j_0$, Lemma 3.1 gives $\mathcal{A} \geq B\ell > 0$. If $\mathcal{A} > 0$ and $0 < B\ell < \mathcal{A}$, the

convergence $F_j \rightarrow \mathcal{A}$ gives j_0 with $F_j \geq B\ell$ for $j \geq j_0$. (b) Integrate $\Pi_j \geq B$ over the window. (c) A strong solution satisfies energy equality, so by (a) it has no floor; if the strong solution from u^0 existed past w , every LH solution from u^0 would coincide with it on $[0, w]$ and have $\mathcal{A}(w_0, w) = 0$. $\square$

No estimate on the nonlinearity is used. Sufficient conditions for energy equality in terms of vanishing shell flux are in [9] for Euler and [10, 34] for Navier-Stokes; the converse required for the equivalence is not in them.

The Floors as Roughness In the language of Sec. 2 the floors are a statement about the regularity of the flow. Energy equality holds on the window whenever $u \in L^3(w_0, w; B_{3,c_0}^{1/3})$ [10, 34], the Navier-Stokes form of the estimate of [9], so $\mathcal{A}(w_0, w) = 0$ on that class and, by Lemma 3.1, $\int_{w_0}^w \Pi_j dt \rightarrow 0$: the class admits no averaged floor, and by Theorem 3.2(a) a positive defect on the window forces $u \notin L^3(w_0, w; B_{3,c_0}^{1/3})$ (from the literature, not formalized). A candidate is therefore Onsager-critical or rougher on the window, over and above the $L_t^2 H_x^1$ regularity the LH class provides; the uniform-in- j floor says that the finest scales carry flux that does not decay, which is what roughness at the exponent $1/3$ means. The converse — a floor from roughness alone — is not in the literature and is not claimed; the equivalence of Theorem 3.2 is with the defect, not with a regularity class.

What the Onsager theory has proved bears on the plausibility of the target and not on the problem itself. For Euler, energy is conserved above the exponent $1/3$ [12, 9] and dissipative Hölder solutions exist below it [21, 4]. For Navier-Stokes the results concern the inviscid limit: Besov regularity above $1/3$, uniform in ν , excludes anomalous dissipation of a family of Leray solutions as $\nu \rightarrow 0$ [13], and forced families whose dissipation stays bounded below uniformly in ν have been constructed [3, 2]. These are statements about the zeroth law distinguished in Sec. 2, a limit along a family, and the constructed members are forced and smooth at each fixed ν , so their defect vanishes; whether one LH solution at one fixed ν has $D(u) \neq 0$ is question (a) of [14] and is unchanged

by them. The floors of this section are the fixed- ν Littlewood-Paley form of that question.

4 What a Positive Defect Must Produce

Set $\Phi(t) = E(t) + \nu \int_0^t \|\nabla u\|^2 ds$, so that $\mathcal{A}(w_0, w) = \Phi(w_0) - \Phi(w)$; Φ is constant for a strong solution, does not increase after any good time, and is lower semicontinuous from the right by weak continuity of u . Let $\Psi(t)$ be the supremum of $\Phi(s)$ over good times $s > t$. The set Σ of singular times is closed, and its $\frac{1}{2}$ -dimensional Hausdorff measure vanishes [30]: covering Σ by intervals, the sum of the square roots of their lengths can be made arbitrarily small. In particular Σ has Hausdorff dimension at most $\frac{1}{2}$ and zero length, so its complement is a countable union of open intervals, on each of which the solution is strong.

Theorem 4.1 (The defect is a jump)

[L: rightProfile_antitone, le_rightProfile, rightProfile_eq_of_mem, apply_le_rightProfile_of_lt, antitone_iff_eq_rightProfile, defect_eq_jump_single_singular_time] (a) Ψ is nonincreasing, $\Phi \leq \Psi$ with equality at good times, and $\Phi(t') \leq \Psi(t)$ for $t < t'$; Φ is nonincreasing iff $\Phi = \Psi$. For good endpoints $\mathcal{A}(w_0, w) = \Psi(w_0) - \Psi(w)$, and Ψ is constant off Σ . (b) Let w_0 be good, w regular, and $T^* \in (w_0, w)$ the only singular time in $[w_0, w]$. Then the one-sided limits $E^\pm(T^*)$ exist, $E(T^*) \leq E^+(T^*) \leq E^-(T^*)$, and

$$\mathcal{A}(w_0, w) = E^-(T^*) - E^+(T^*) ,$$

$$\mathcal{A}(w_0, T^*) = E^-(T^*) - E(T^*) \geq \mathcal{A}(w_0, w) .$$

Proof. (a) Good times are dense, so Ψ is a supremum over a nonempty set bounded by Φ at any earlier good time, shrinking as t grows. Right lower semicontinuity gives, for any t and $\varepsilon > 0$, a good $s > t$ with $\Phi(t) \leq \Phi(s) + \varepsilon \leq \Psi(t) + \varepsilon$; at a good t every later $\Phi(s)$ is at most $\Phi(t)$; for $t < t'$ a good $s \in (t, t')$ gives $\Phi(t') \leq \Phi(s) \leq \Psi(t)$. (b) On (w_0, T^*) and $(T^*, w]$ the solution is strong and Φ takes constant values $L = \Phi(w_0)$ and $R = \Psi(T^*)$; the dissipation integral

is continuous, so $E^\pm = L, R - \nu \int_0^{T^*} \|\nabla u\|^2$ exist, and weak lower semicontinuity gives $\Phi(T^*) \leq R \leq L$. $\square$

A positive defect across a single singular time is a jump of the kinetic energy: the solution converges weakly in L^2 from both sides to $u(T^*)$, and the energy it carries drops. The window may also end at T^* with $E(T^*) < E^+(T^*) = E^-(T^*)$, which the LH axioms do not exclude; then the floors of Theorem 3.2 exist only on windows ending at T^* . A proof must produce energy retained as $t \uparrow T^*$, a lower bound where energy methods give upper bounds.

Where the Energy Goes For a suitable weak solution the inertial dissipation of [14] is a nonnegative measure on space-time (Proposition 1 there with the local energy inequality of [5]), supported on the singular set, which has parabolic one-dimensional measure zero [5], with mass $\mathcal{A}(w_0, w)$ on $\mathbb{T}^3 \times (w_0, w]$. At a single singular time the lost energy is carried by the slice of the singular set at that time, a set of zero length, the energy measure of [23]. That measure has an *atom* at a point x_0 if the single point $\{x_0\}$ carries positive mass, and its *atomic part* is the sum of the masses of its atoms, necessarily countably many; the jump has an atomic part when some single point carries a positive share of the energy lost at T^* , and is atomless when the loss spreads over the slice with no point carrying any of it on its own. Six facts restrict the singularity that carries the jump.

1. A Leray backward self-similar solution cannot carry it: such solutions do not exist [25, 36], and formally the profile $(T^* - t)^{-1/2} U(x/\sqrt{T^* - t})$, $U \in L^2$, carries energy $(T^* - t)^{1/2} \|U\|^2 \rightarrow 0$ into the singular point.
2. A blowup of Type I in time, $\|u(t)\|_\infty \leq C(T^* - t)^{-1/2}$, satisfies energy equality across the first singular time [23, 11].
3. If the local energy in balls of radius r about the singular slice is at most Cr for times within r^2 of T^* , a finite cover of the zero-length slice of total radius ε holds at most $C\varepsilon$ of energy near T^* , and the rest converges: no jump [L: no_jump_of_small_covers]. The bound need not be uniform (argued,

not formalized): the points where it holds with constant N below radius $1/N$ carry no excess, so the jump is carried by points where the local energy exceeds every multiple of r .

4. For the Euler equations a bounded gradient rate $(T^* - t)\|\nabla u(t)\|_\infty \leq K$ excludes atoms of the energy measure [6]. The argument transfers to the smooth branch of an LH solution before its first singular time (argued, not formalized): the energy-conserving zoom $u = r^{-3/2}v((x - x_0)/r, T^* + r^{5/2}s)$ of [31] carries the L^2 mass and the gradient rate unchanged and multiplies the viscosity by $r^{1/2}$; the limit is an Euler flow on $(-1, 0)$ with a Dirac endpoint measure, which the Liouville theorem of [6] annihilates, given Lagrangian transport of vorticity supports for Lipschitz weak Euler solutions ([6] establish that transport for smooth solutions; the Lipschitz weak case is proved separately). Hence an atom of the energy measure at T^* , in particular a jump carried by a countable slice, forces $\limsup_{t \uparrow T^*} (T^* - t)\|\nabla u(t)\|_\infty = \infty$. An atomless excess is not excluded.
5. The pressure leaves L^2 . If $p - \beta(t) \in L^2(\mathbb{T}^3 \times (\tau, T^*))$ for some β and some $\tau < T^*$, the energy is continuous at T^* [22]; with D the left jump of $\|u\|^2$ at T^* , $\inf_\beta \|p - \beta\|_{L^2(\mathbb{T}^3 \times (\tau, T^*))} \geq D\sqrt{\nu}/(\sqrt{2}\|u(\tau)\|)$ for every $\tau < T^*$, and an atom at x_0 forces the same failure on every terminal neighbourhood of x_0 [20]. Since $u \in L_t^4 L_x^4$ implies $p \in L_{t,x}^2$, this moves the exit from $L_t^4 L_x^4$ [24, 33] to the pressure; dimensionally it is the same condition.
6. The scale of the concentration is constrained twice. (a) Let $\lambda(t)$ be the least r with $\int_{B_r(x_0)} |u(t)|^2 \geq m_0/2$ for an atom of mass m_0 at x_0 . The Poincaré inequality on a ball of radius proportional to $\lambda(t)$ gives $\|\nabla u(t)\|^2 \geq cm_0^{5/3} E_0^{-2/3} \lambda(t)^{-2}$ (argued, not formalized), and the energy inequality then bounds $\int^{T^*} \lambda(t)^{-2} dt \leq CE_0^{5/3}/(\nu m_0^{5/3})$ [L: superdiffusive_integral_bound]; so $\lambda(t)^2 \leq K\nu(T^* - t)$ holds on no terminal interval [L: not_

`diffusive_on_terminal_interval`]: a collapse whose rate is selected by viscosity carries no atom. (b) A collapse onto a fixed profile, $u \approx a\lambda^{-3/2}U_0((x-x_0)/\lambda)$ with U_0 a Lipschitz finite-energy steady Euler flow, has first-order corrector equation $L_0T_1 = -\Lambda U_0$, $\Lambda = \frac{3}{2} + y \cdot \nabla$, which requires ΛU_0 orthogonal to the kernel of the adjoint linearized Euler operator [L: `not_mem_range_of_adjoint_kernel_pairing`]; for axisymmetric flows the kernel element $h(\psi)Re_\phi$ gives the swirl condition $\frac{1}{2}FV' + 3F'V = 0$ across the levels of the stream function, which makes $FV^{1/6}$ constant, hence $F \equiv 0$ for compact support [L: `swirl_const_of_C3`, `swirl_zero_of_C3_boundary`].

A positive defect therefore requires a Type II energy-concentrating singularity of Navier-Stokes on $\mathbb{T}^3$ at fixed viscosity: blowup of Type II in velocity, and in gradient rate if the jump has an atomic part; a fixed amount of L^2 energy concentrating onto a set of zero length as $t \uparrow T^*$, some point of which carries local energy exceeding every multiple of its radius; a pressure outside L^2 on every terminal interval; and, by Sec. 3, a velocity outside the Onsager-critical class $L_t^3 B_{3,c_0}^{1/3}$ on the window. These conditions are necessary, not sufficient.

A Designed Model Without a Defect Tao's averaged Navier-Stokes equation [35] has the energy identity and the harmonic-analysis estimates of the true equation and a proven finite time blowup in the LH class, of Type II by scaling. Its defect up to T^* is zero (argued from the bounds of [35], not formalized): the auxiliary modes that time its designed cascade retain a fixed fraction of each stage's energy, so the front carries vanishing energy and $E(t)$ is continuous at T^* from the left, $\mathcal{A}(w_0, T^*) = 0$ for every w_0 . Blowup, Type II, exit from $L_t^4 L_x^4$ and a cascade faster than dissipation hold with zero defect: a positive defect requires asymptotically complete transfer to fine scales. No autonomous energy-conserving system with the Navier-Stokes estimates is known to satisfy the analogue of a positive defect.

The Announced Constructions Against These Conditions The announced Navier-Stokes solution

[27] is forced, with $u(0) = 0$ and $f \in C_c^\infty(\mathbb{R}^3 \times (0, \infty))$, and lies outside the class in which the positive defect problem is posed; its energy identity carries the input $\langle f, u \rangle$ (Lemma 10.4 there). Its velocity scale is $(1-t)^{-1/2-h}$ with $0 < h < 1/100$, so the blowup is of Type II in velocity; the kinetic energy of the collapsing core is of order $(1-t)^{1/2-3h}$ and tends to zero, and the dissipation integral up to the singular time is finite (Secs. 2.1, 3.5, Lemma 10.4 there). The core carries no energy atom; the energy at the singular time, the energy inequality across it and the inertial dissipation of [14] are not discussed. The solution is not a candidate and is not presented as one. The Alpöge-Buckmaster solution is a forced Euler flow and the Euler solution of [28] is inviscid; neither concerns the equation of Sec. 2.

5 What Cannot Certify a Positive Defect

Computation No finite computation witnesses a positive defect, and the obstruction is structural rather than a matter of resolution or of computer time: every object a computation produces lives where the defect is zero. A Galerkin truncation obeys an exact energy identity, so its own defect vanishes, and an enclosure of the true solution is valid only while that solution is strong — where the defect vanishes as well. The rest of this paragraph makes the statement precise for validated computation: the bridge that turns a computed trajectory into a bound on the epoch of regularity, the theorem that the bridge cannot close on a trajectory that also respects the ceiling (Theorem 5.1), and the fact that every ceiling now known is inherited by the truncation with the same constant.

In validated computation [8] a Galerkin truncation is computed with interval arithmetic and an a posteriori bound encloses the true solution in a tube around the computed one while the true solution is regular [L: `aposteriori_gronwall`]. The bridge [L: `validated_epoch_ends_inside`] turns verified floors B and one analytic ceiling Ct^{-q} into $e \leq \max(w_0, (C/(B-\varepsilon))^{1/q})$, ε the enclosure width; the argument closes when this

bound lies below w .

The rate here is not ν_t , for the reason given in Sec. 2: a Galerkin truncation satisfies the energy identity exactly, so its inertial dissipation rate vanishes identically and no positive floor on it is verifiable. What the computation has is the flux out of the resolved shells of the computed trajectory, the observable of Sec. 3 evaluated there; write it $\tilde{\nu}_t$. The theorem below is a statement about $\tilde{\nu}_t$.

Theorem 5.1 (Galerkin-uniform ceilings are unwitnessable) [L: `galerkin.uniform.ceiling.unwitnessable`, `witness.requires.truncation.violation`]

Let the computed trajectory $\tilde{\nu}_t$ carry the verified floors $B \leq \tilde{\nu}_t$ on $[w_0, w)$ and also obey the ceiling $\tilde{\nu}_t \leq Ct^{-q}$ there. Then for every enclosure $0 \leq \varepsilon < B$ the bridge does not close: $\max(w_0, (C/(B - \varepsilon))^{1/q}) \geq w$. Equivalently, if it closes, the computed trajectory exceeds the ceiling somewhere on the window.

Proof. At any t in the window, $Bt^q \leq C$. The margin check requires $(B - \varepsilon)t^q > C$ for t near w . Together, $\varepsilon t^q < 0$, a contradiction for $\varepsilon \geq 0$. $\square$

Every known ceiling is proved by energy methods that commute with the Galerkin projection and so holds for the truncated system with the same constant: the energy budget, the $L^{4/3}$ bounds of the Leray-Hopf class, and the analytic ceiling Ct^{-q} from Gevrey class regularity [18]. Whether a ceiling on regular flows exists that Galerkin truncations violate is open. Computation can locate a configuration and certify finitely many inequalities inside a proof [8]; it cannot witness a positive defect.

Depth vs. Timing The averaged floor requires two things of the energy transfer at once. The transfer must persist in time: it must stay above a fixed level across a window of positive length, not merely at a single instant. And it must persist in scale: the same level, with the same constant, at every scale finer than some fixed one, however fine — in the coarse-graining language of Sec. 3, a floor on the flux Π_ℓ through every length scale ℓ below some ℓ_0 , not only through those a given resolution resolves. The

first requirement is that the cascade be sustained, the second that it continue to arbitrarily fine scales.

The two are not equally difficult, and ordinary turbulence shows the difference. Eddies of neighbouring size transfer energy to one another after a short delay, so the flux through a band of fine scales rises and falls almost simultaneously, the lag from one octave to the next being a small fraction of the turnover time of the large eddies. A floor common to such a band therefore holds, and holds across a window comparable with the turnover time rather than at one instant. The second requirement is not met, for the reason the phenomenology already gives: at a fixed viscosity the cascade ends at the Kolmogorov scale, viscous dissipation removes the whole of the flux arriving there, and below that scale no transfer remains.

As a concrete example, the authors computed decaying turbulence on the periodic cube from random large-scale data over a range of viscosities and recorded the transfer out of each dyadic band at every step. Against the time condition the computed flow attains, to within a few percent, the highest common floor its own history admits; the fine bands take their peak transfer within about a third of a turnover time of one another, and averaging over a window of that length lowers the floor only slightly. Against the scale condition the common floor falls with every octave added towards the Kolmogorov scale, and at the last band before that scale the transfer is smaller than at the large scales by more than an order of magnitude. These are floating-point simulations, not the validated computations of the preceding paragraph: they measure how far a computed flow is from each of the two conditions and establish nothing about the true solution.

Two statements behind this reading are machine-checked: no computed flow can be certified to hold a floor above the ceiling that its own history defines [L: `certified.profile.le.one`, `ceiling.dominates.weighted.sup`], and in a model where each band repeats a common profile after a delay, the shortfall is determined by the spread of the peak times alone [L: `lag.model.margin.bound`, `lag.model.crossing.iff.simultaneous`]. The second is why the time condition is decided by a single quantity.

Neither a finer grid nor a higher Reynolds number

changes the second condition. The Kolmogorov scale is set by the viscosity and the dissipation rate, not by the resolution, so refining the grid resolves the viscous range in more detail without extending the range of scales over which a floor could hold. Nor can the reduction weaken that condition: by Theorem 3.2(c), a floor holding past the Kolmogorov scale at fixed viscosity already implies that the flow is not strong on the window, which is the conclusion sought. A candidate for a positive defect must therefore differ from generic turbulence in the depth of its cascade, and not in its timing.

Selection The maximum rate of entropy production [19, 7] selects, among the evolutions consistent with the constraints, the one that degrades energy fastest; its well-posed form is extremal selection on a compact set of solutions with $\mathcal{A}$ upper semicontinuous. A defect-maximizing solution exists, and either every solution in the set has $\mathcal{A} \leq 0$ or the maximizer has a positive defect [L: `extremal_defect_exists`, `defect_dichotomy`]. Selection locates a solution if one exists; it cannot create one.

Forcing With a force at large scales and a stationary statistical solution [17], uniform mean flux floors at all fine scales hold iff the mean defect is positive [L: `stationary_mean_flux_ge_defect`, `stationary_mean_flux_tendsto_defect`]: for the floor hypothesis, forcing changes nothing at fixed viscosity. Whether a force can produce a singularity is a separate question (Sec. 4).

6 Closing

Integrated over its window, the hypothesis of the floor-ceiling argument is a positive defect; at the level of the shells that defect is exactly an averaged flux floor, which the pointwise floor implies. A proof must produce a Type II energy-concentrating singularity; no computation or selection principle supplies one, and forcing does not remove the need for a positive defect; the authors' search shows generic turbulence meeting the time condition of the reduction and failing the scale condition at the Kolmogorov scale. On

the information available at the time of writing, the forced constructions of [27, 1] leave the positive defect problem open, as they leave the unforced questions (A) and (B); the conditions of this paper are the admissibility criteria for a programmatic search that approaches those questions through a positive defect.

7 Acknowledgements

The authors thank Abdul Hasib Rahimyar for his reading of the paper and his comments, which prompted the passages relating the shells and the flux to Littlewood-Paley theory, coarse-graining and the Onsager theory of turbulence in Secs. 2 and 3. The machine verification, the archived computations, and the reduction reported here were carried out with the assistance of an AI research agent; every Lean declaration compiles independently of that assistance and may be checked by the reader.

References

- [1] L. ALPÖGE AND T. BUCKMASTER, *Blowup for the Euler equations with smooth forcing*. preprint dated 7 September 2026, <https://cims.nyu.edu/~tristanb/euler.pdf>, 2026. No author line; authors as named in the OpenAI announcement of 8 September 2026. Accessed 9 September 2026.
- [2] E. BRUÈ, M. COLOMBO, G. CRIPPA, C. DE LELLIS, AND M. SORELLA, *Onsager critical solutions of the forced Navier-Stokes equations*, Commun. Pure Appl. Anal., 23 (2024), pp. 1350–1366.
- [3] E. BRUÈ AND C. DE LELLIS, *Anomalous dissipation for the forced 3D Navier-Stokes equations*, Comm. Math. Phys., 400 (2023), pp. 1507–1533.
- [4] T. BUCKMASTER, C. DE LELLIS, L. SZÉKELYHIDI, JR., AND V. VICOL, *Onsager's conjecture for admissible weak solutions*, Comm. Pure Appl. Math., 72 (2019), pp. 229–274.
- [5] L. CAFFARELLI, R. KOHN, AND L. NIRENBERG, *Partial regularity of suitable weak solutions of the Navier-Stokes equations*, Comm. Pure Appl. Math., 35 (1982), pp. 771–831.
- [6] D. CHAE AND J. WOLF, *Energy concentrations and Type I blow-up for the 3D Euler equations*, Comm. Math. Phys., 376 (2020). arXiv:1706.02020.
- [7] G.-Q. CHEN, J. GLIMM, AND H. SAID, *A principle of maximum entropy for solutions for the Navier-Stokes equation*, Physica D, 467 (2024).

- [8] J. CHEN AND T. HOU, *Stable nearly self-similar blowup of the 2d Boussinesque and 3d Euler equations with smooth data ii: Rigorous numerics*, arXiv, (2023). arXiv:2305.05660v1 [math.AP].
- [9] A. CHESKIDOV, P. CONSTANTIN, S. FRIEDLANDER, AND R. SHVYDKOY, *Energy conservation and Onsager's conjecture for the Euler equations*, Nonlinearity, 21 (2008), pp. 1233–1252.
- [10] A. CHESKIDOV, S. FRIEDLANDER, AND R. SHVYDKOY, *On the energy equality for weak solutions of the 3D Navier-Stokes equations*, in Advances in Mathematical Fluid Mechanics, Springer, 2010, pp. 171–175.
- [11] A. CHESKIDOV AND X. LUO, *Energy equality for the Navier-Stokes equations in weak-in-time Onsager spaces*, Nonlinearity, 33 (2020), pp. 1388–1403.
- [12] P. CONSTANTIN, W. E, AND E. S. TITI, *Onsager's conjecture on the energy conservation for solutions of Euler's equation*, Comm. Math. Phys., 165 (1994), pp. 207–209.
- [13] T. D. DRIVAS AND G. L. EYINK, *An Onsager singularity theorem for Leray solutions of incompressible Navier-Stokes*, Nonlinearity, 32 (2019), pp. 4465–4482.
- [14] J. DUCHON AND R. ROBERT, *Inertial energy dissipation for weak solutions of incompressible Euler and Navier-Stokes equations*, Nonlinearity, 13 (2000), pp. 249–255.
- [15] G. L. EYINK, *Energy dissipation without viscosity in ideal hydrodynamics. I. Fourier analysis and local energy transfer*, Physica D, 78 (1994), pp. 222–240.
- [16] C. L. FEFFERMAN, *Existence and smoothness of the Navier-Stokes equation*, in The Millennium Prize Problems, J. Carlson, A. Jaffe, and A. Wiles, eds., Clay Math. Inst. and Amer. Math. Soc., 2006, pp. 57–67. Official problem statement, <https://www.claymath.org/wp-content/uploads/2022/06/navierstokes.pdf>.
- [17] C. FOIAS, O. MANLEY, R. ROSA, AND R. TEMAM, *Navier-Stokes Equations and Turbulence*, Cambridge University Press, Cambridge, 2004.
- [18] C. FOIAS AND R. TEMAM, *Gevrey class regularity for the solutions of the Navier-Stokes equations*, J. Funct. Anal., 87 (1989), pp. 359–369.
- [19] J. GLIMM, D. LAZAREV, AND G.-Q. CHEN, *Maximum entropy production rate as a necessary admissibility condition for the fluid Euler and Navier-Stokes equations*, SN Applied Sciences, 2 (2020), p. 160. <https://doi.org/10.1007/s42452-020-03941-2>.
- [20] H. HUANG, *Endpoint energy atoms force local pressure concentration in three-dimensional Navier-Stokes flow*. arXiv:2608.30715, 2026.
- [21] P. ISETT, *A proof of Onsager's conjecture*, Ann. of Math. (2), 188 (2018), pp. 871–963.
- [22] I. KUKAVICA, *Role of the pressure for validity of the energy equality for solutions of the Navier-Stokes equation*, J. Dynam. Differential Equations, 18 (2006), pp. 461–482.
- [23] T. M. LESLIE AND R. SHVYDKOY, *The energy measure for the Euler and Navier-Stokes equations*, Arch. Ration. Mech. Anal., 230 (2018), pp. 459–492.
- [24] J.-L. LIONS, *Sur la régularité et l'unicité des solutions turbulentes des équations de Navier Stokes*, Rend. Sem. Mat. Univ. Padova, 30 (1960), pp. 16–23.
- [25] J. NEČAS, M. RŮŽIČKA, AND V. ŠVERÁK, *On Leray's self-similar solutions of the Navier-Stokes equations*, Acta Math., 176 (1996), pp. 283–294.
- [26] L. ONSAGER, *Statistical hydrodynamics*, Nuovo Cimento (9), 6 (1949), pp. 279–287.
- [27] OPENAI, *Finite time blowup for Navier-Stokes*. <https://cdn.openai.com/pdf/32d9f210-8b73-45e0-91bc-82a30aef8a9a/navier-stokes.pdf>; announcement <https://openai.com/index/navier-stokes-solution/>; Lean formalization <https://github.com/openai/NavierStokesAndEuler>, 2026. Accessed 8–9 September 2026.
- [28] ———, *Finite time blowup for the Euler equation*. <https://cdn.openai.com/pdf/315b36cd-ec98-4023-8342-93345194ece1/euler.pdf>, 2026. Accessed 8 September 2026.
- [29] G. PRODI, *Un teorema di unicità per le equazioni di Navier-Stokes*, Ann. Mat. Pura Appl., 48 (1959), pp. 173–182.
- [30] J. ROBINSON, J. RODRIGO, AND W. SADOWSKI, *The Three-Dimensional Navier-Stokes Equations*, Cambridge University Press, Cambridge, 2016.
- [31] G. SEREGIN, *Remarks on Type II blowups of solutions to the Navier-Stokes equations*. arXiv:2304.04045; Commun. Pure Appl. Anal. (2024), 2024.
- [32] J. SERRIN, *On the interior regularity of weak solutions of the Navier-Stokes equations*, Arch. Ration. Mech. Anal., 9 (1962), pp. 187–195.
- [33] M. SHINBROT, *The energy equation for the Navier-Stokes system*, SIAM J. Math. Anal., 5 (1974), pp. 948–954.
- [34] R. SHVYDKOY, *Lectures on the Onsager conjecture*, Discrete Contin. Dyn. Syst. Ser. S, 3 (2010), pp. 473–496.
- [35] T. TAO, *Finite time blowup for an averaged three-dimensional Navier-Stokes equation*, J. Amer. Math. Soc., 29 (2016), pp. 601–674.
- [36] T.-P. TSAI, *On Leray's self-similar solutions of the Navier-Stokes equations satisfying local energy estimates*, Arch. Ration. Mech. Anal., 143 (1998), pp. 29–51.
- [37] T. UDA, *Lean 4 formalization of Leray-Hopf weak solution existence for the three-dimensional incompressible Navier-Stokes equations*. <https://github.com/uda-lab/leray-hopf>, 2026.